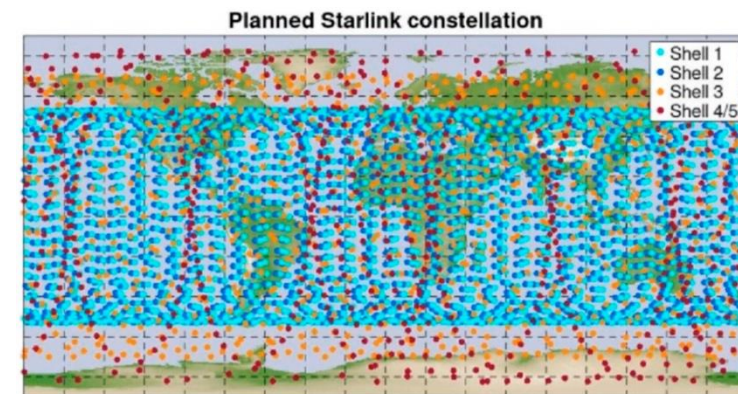


Simulation of Passive Synthetic Aperture Radar for Starlink

Rawsthorne Grundy

Introduction and Motivation

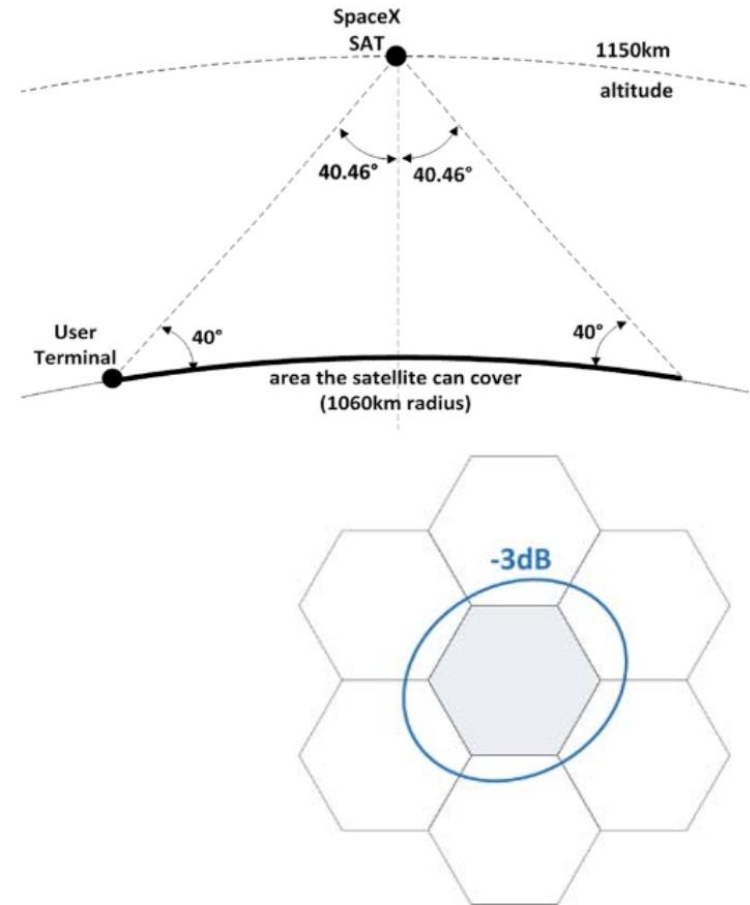
- Passive Radar From Satellites (GNSS)
- Growing LEO Satellite Constellations
- 10,000 Starlink Satellites and Growing
- Goals
 - Explore Passive SAR Modalities
 - Develop a Tool For Producing Starlink Downlink Waveforms
 - Simulate SAR Using Starlink Waveforms



Blázquez-García, R., et al.: Capabilities and challenges of passive radar systems based on broadband low-Earth orbit communication satellites. IET Radar Sonar Navig. 18(1), 78–92 (2024). <https://doi.org/10.1049/rsn2.12446>

Downlink

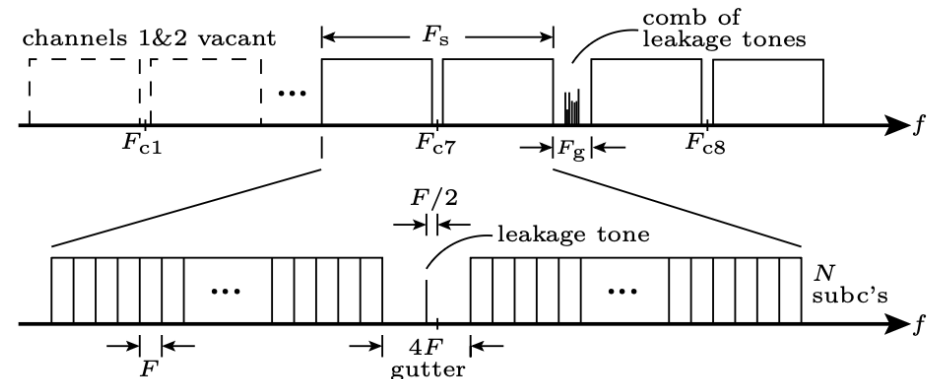
- Communication Between the Satellite and User Terminals
- Consistent, Earth Pointed Signal
- Prevalence Depends on User Need
- Constantly Moving Signal Source
- Illuminator of Opportunity



A. Sayin, M. Cherniakov and M. Antoniou, "Passive radar using Starlink transmissions: A theoretical study," *2019 20th International Radar Symposium (IRS)*, Ulm, Germany, 2019, pp. 1-7, doi: 10.23919/IRS.2019.8768105.

Frequency Structure

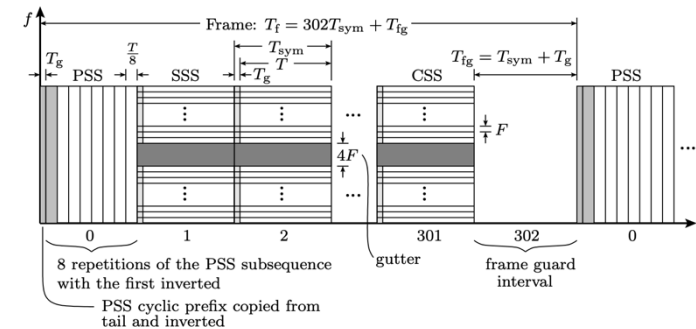
- 8 Ku-Band Channels spread from 10.7 GHz to 12.7 GHz
 - Channels 1 and 2 are vacant
- 240 MHz Bandwidth per Channel
- 1024 Subcarriers per Channel with a 4 Nulls in the Center



T. E. Humphreys, P. A. Iannucci, Z. M. Komodromos and A. M. Graff, "Signal Structure of the Starlink Ku-Band Downlink," in *IEEE Transactions on Aerospace and Electronic Systems*, vol. 59, no. 5, pp. 6016-6030, Oct. 2023, doi: 10.1109/TAES.2023.3268610.

Frame Encoding

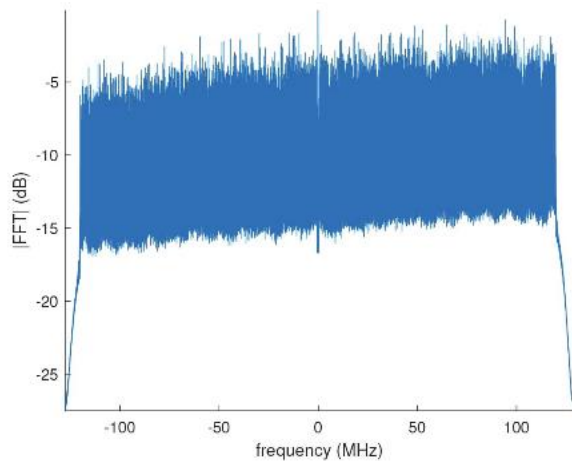
- 1/750 s Long Frame
- PSS
 - 8 Repetitions of DPSK Encoded M-Sequence
 - First Repetition Inverted
- SSS
 - 4QAM OFDM Encoded Combined M-Sequences
- Header
 - 3 Symbols of Position Data
- Payload
 - 16QAM and 4QAM
- CSS and CM1SS
- Guard Between Frames



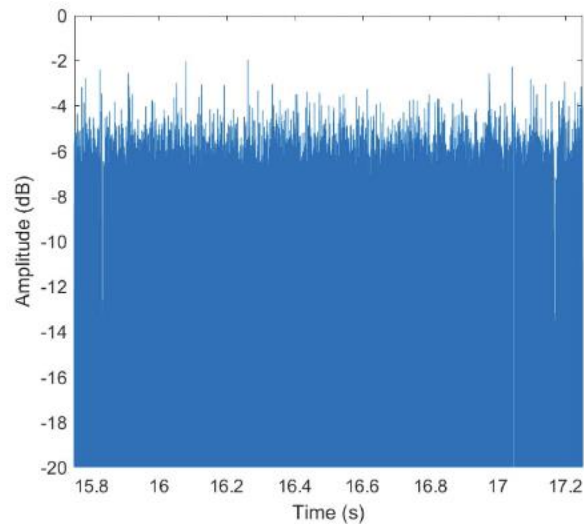
T. E. Humphreys, P. A. Iannucci, Z. M. Komodromos and A. M. Graff, "Signal Structure of the Starlink Ku-Band Downlink," in *IEEE Transactions on Aerospace and Electronic Systems*, vol. 59, no. 5, pp. 6016-6030, Oct. 2023, doi: 10.1109/TAES.2023.3268610.

Recorded Data From Literature

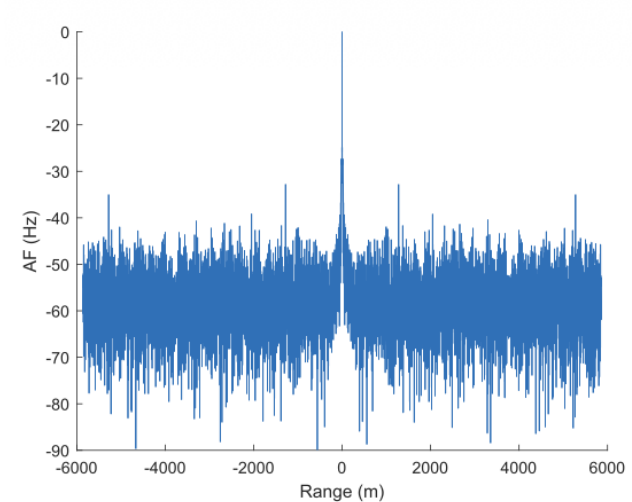
Spectrum



Time Series



Autocorrelation



Gomez-del-Hoyo, P. and Samczynski, P., "Starlink-Based Passive Radar for Earth's Surface Imaging: First Experimental Results", IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, vol. 17, IEEE, pp. 13949–13965, 2024.
doi:10.1109/JSTARS.2024.3437179.

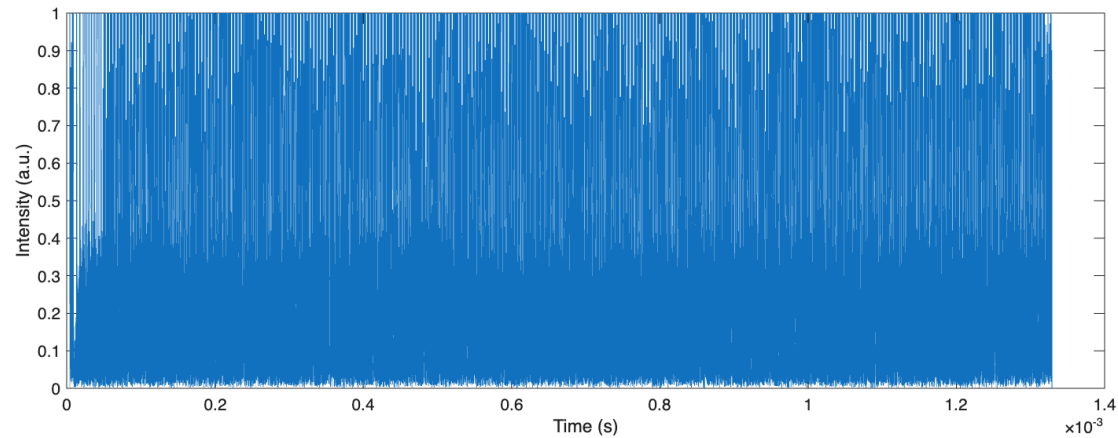
Signal Re-Creation

- Created MATLAB program that generates a signal encoding of an arbitrary file.
- Uses known hex values of PSS and SSS sequences from Humphreys et al.
- Draws header info from arbitrary text file
- Random CSS and CM1SS data but constant from frame to frame
- Normalized output with OFDM peak clipping for power consistency

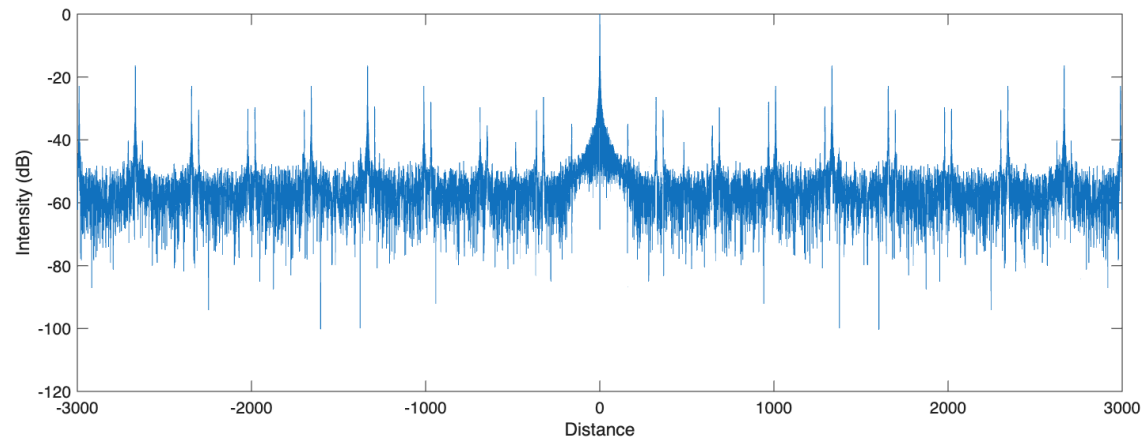
Signal Result

Time Series

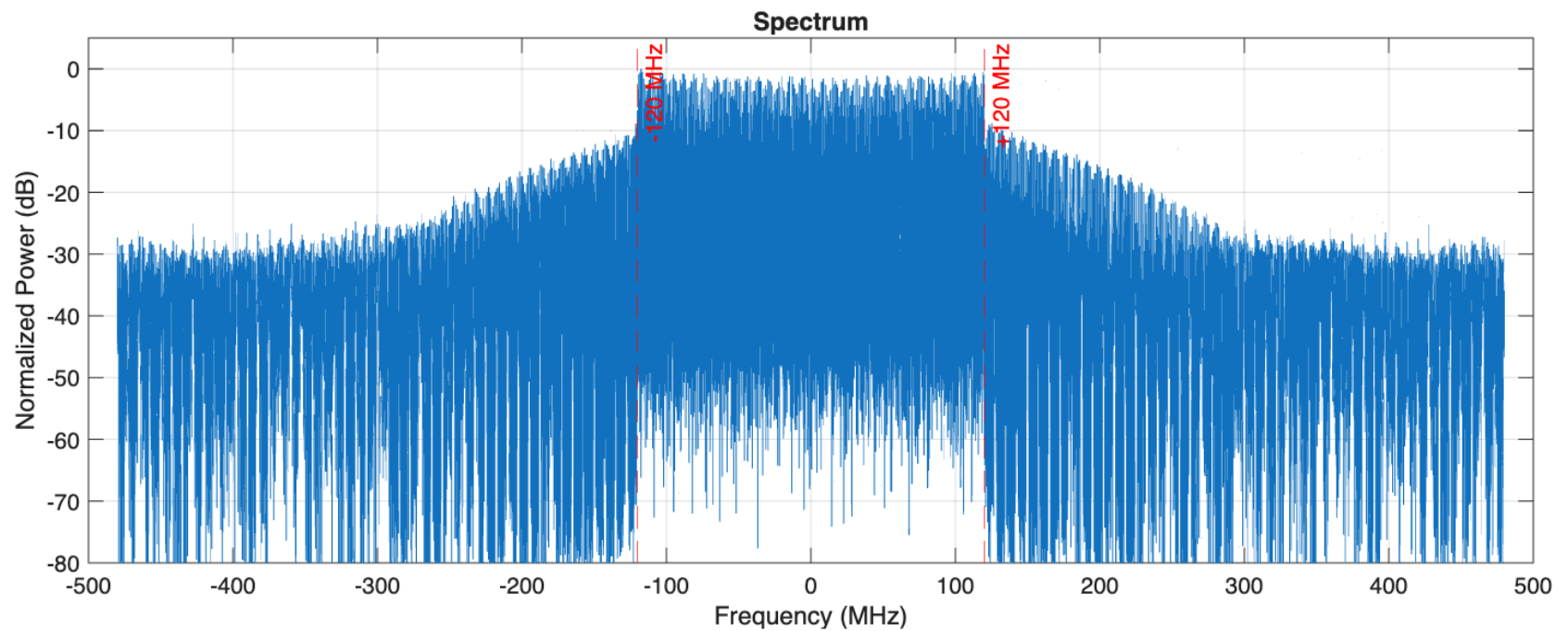
Clipped at 5x rms



Autocorrelation

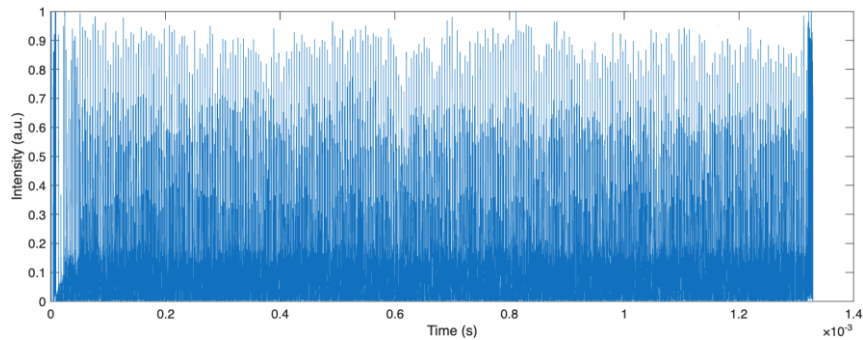


Signal Spectrum

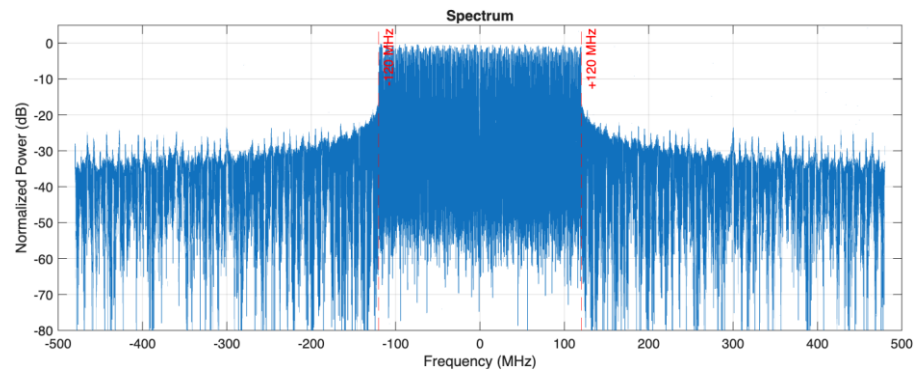
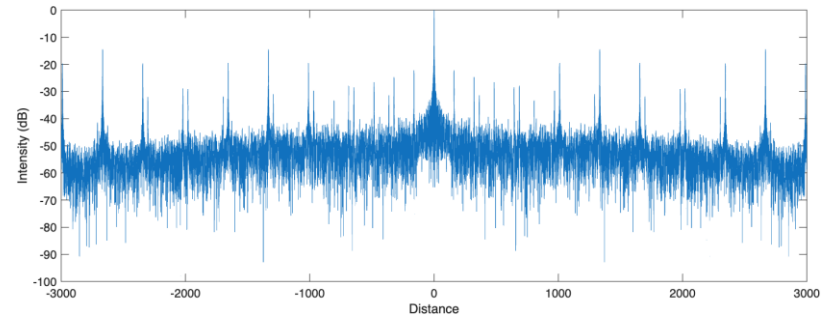


Signal Spectrum with Varied Clipping

Time Series



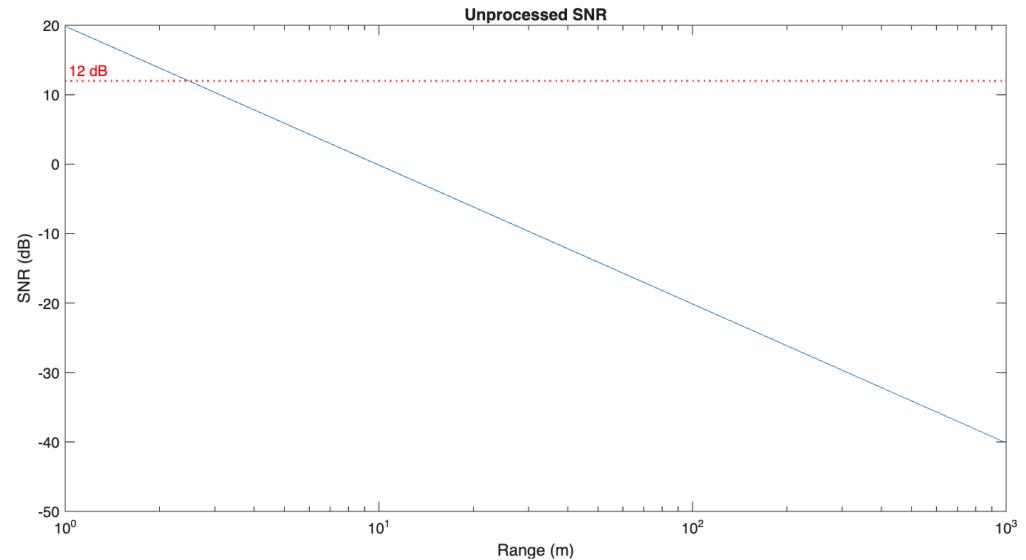
Autocorrelation



Clipped at 1000x
RMS

Radar System Characteristics

$$\begin{aligned}\lambda &= \frac{c}{\nu} = .023m = 23mm \\ R_{tx} &= 500 km \\ G_{tx} &= 34.0 dB = 2511.8 \\ EIRP &= 45.1 dB \\ P_T &= 10^{(EIRP-G)/10} = 12.882 W \\ R_{rx} &= 1000m \\ \sigma &= 100 m^2 \\ G_{rx} &= 20 dB = 100 \\ L &= 6 dB \\ B &= 240 MHz \\ T &= 290 K\end{aligned}$$



$$\begin{aligned}P_R = \text{Received Signal} &= P_T G_{tx} \frac{1}{4 \pi R_{tx}^2} \frac{1}{L} \sigma \frac{1}{4 \pi R_{rx}^2} \frac{G_{rx} \lambda^2}{4 \pi} = -160.6 dBW \\ N &= kTB = 960 * 10^{-15} W = -120.17 dBW \\ SNR &= P_R - N = -40.42 dBW\end{aligned}$$

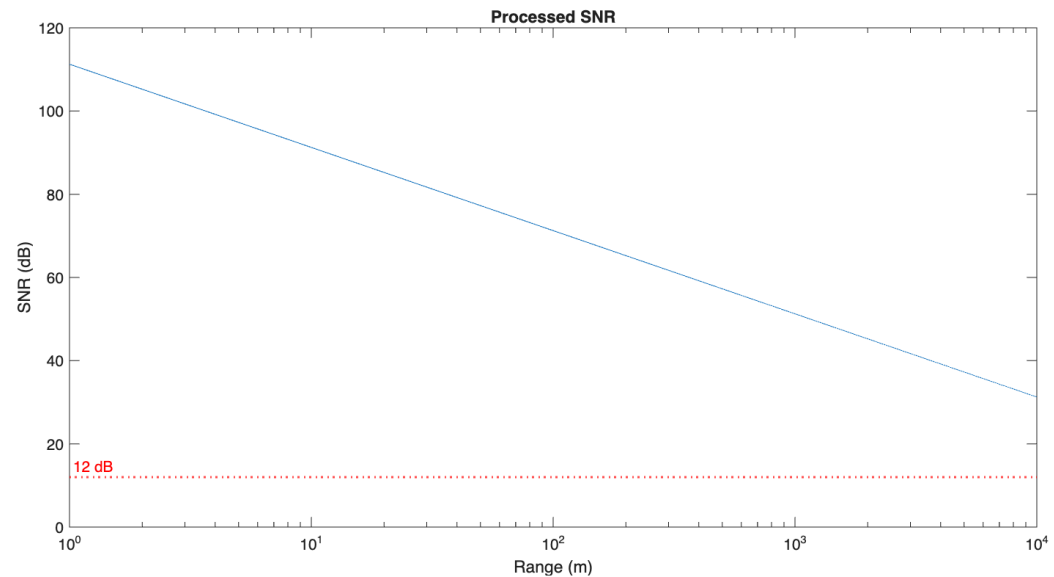
Processing Gain

■ Pulse Compression Through Matched Filtering

- $Compression\ Gain = 2B\tau = 2(240MHz) \left(\frac{1}{750} s \right) = 55\text{ dB}$

■ Dwell Time

- 2.5 degree 3dB antenna beamwidth
- 500 km Elevation
- 44 km Spot Size
- 7.5 km/s Satellite Velocity
- 5.8 s or 4366 pulses spent on target
- Coherent Integration Gain of 36.4 dB



Resolution Estimates

$$v = 7.5 \text{ km/s}$$

$$R = 500 \text{ km}$$

$$\lambda = 23 \text{ mm}$$

$$T_a = 4 \text{ s}$$

$$\Delta CR = \frac{\lambda R}{2 v T_a} = 17.5 \text{ cm}$$

$$BW = 240 \text{ MHz}$$

$$\text{Range Resolution} = \frac{c}{2BW} = .625 \text{ m}$$

Simulation

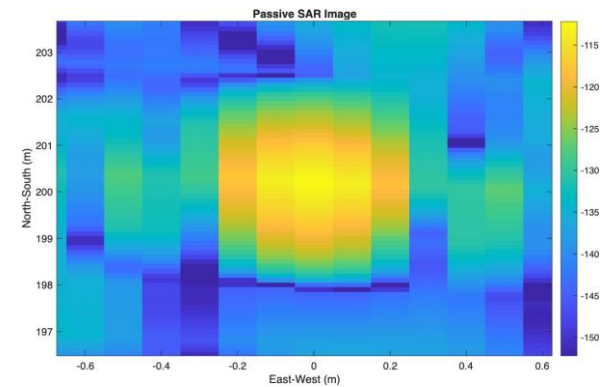
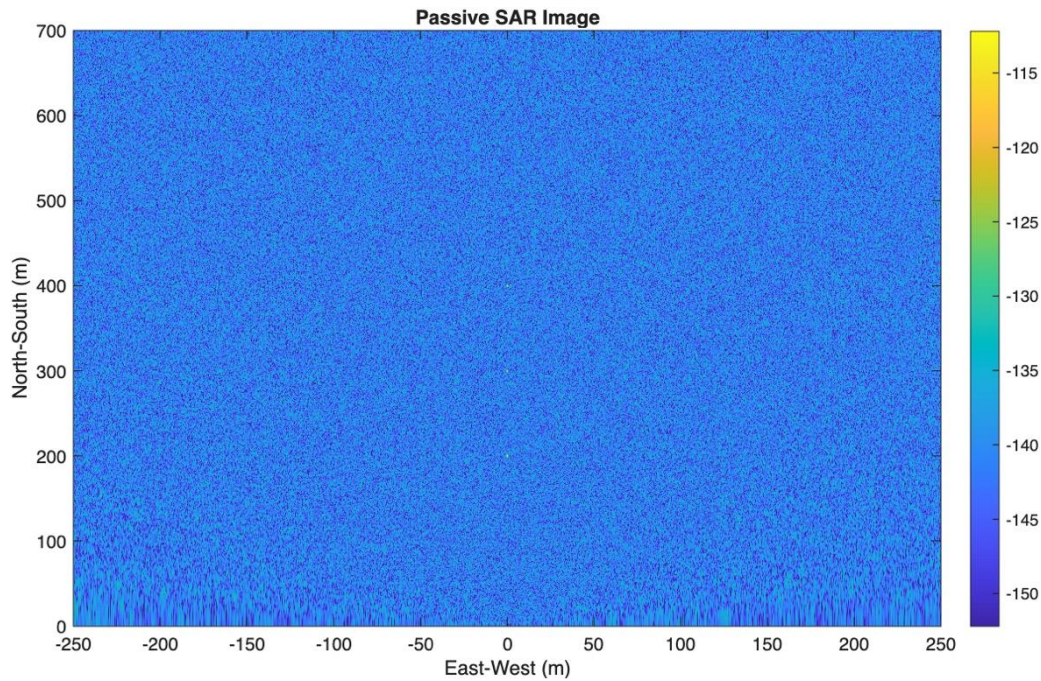
- MATLAB Radar Toolbox
- Independent Propagation Pathways
- Assumptions
 - Straight Satellite Path
 - Flat Earth
 - Isotropic Emitters
 - Constant RCS
- Satellite at [-25000 m, 50,000 m , 0 m] with a Velocity of [7500 m/s, 0m/s, 0 m/s]
- Point Targets at [0, 200 m, 0], [0 m , 300 m, 0 m], and [0 m, 400 m, 0 m]

Processing

- **Matched Filtering of the Reference and Signal Channel for Differential Range**
 - Removes the need for doppler compensation.
 - Allows use of full pulse length
- **Back Projection for Azimuth Compression**
 - Uses known transmit positions and interpolated range profiles from matched filtering step
 - Best method given the variable flight path of the satellite and complexity.

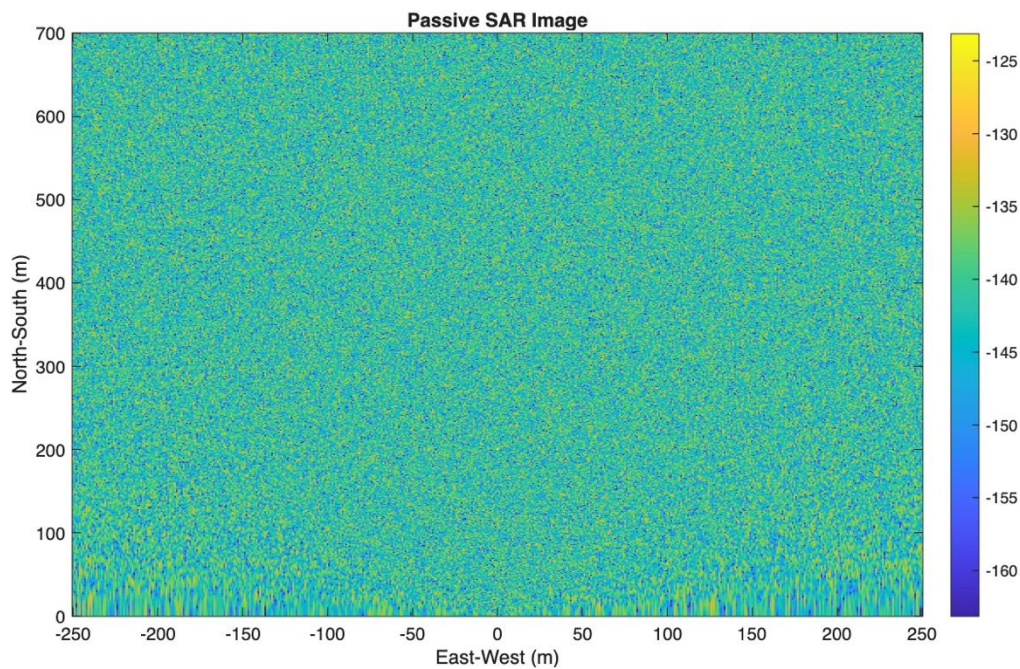
Results

5000 Pulses
Target RCS of 1000 m²

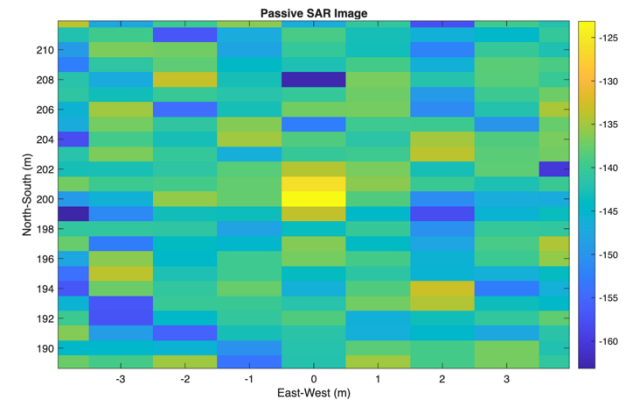


Results Continued

5000 Pulses
Target RCS of 100 m^2



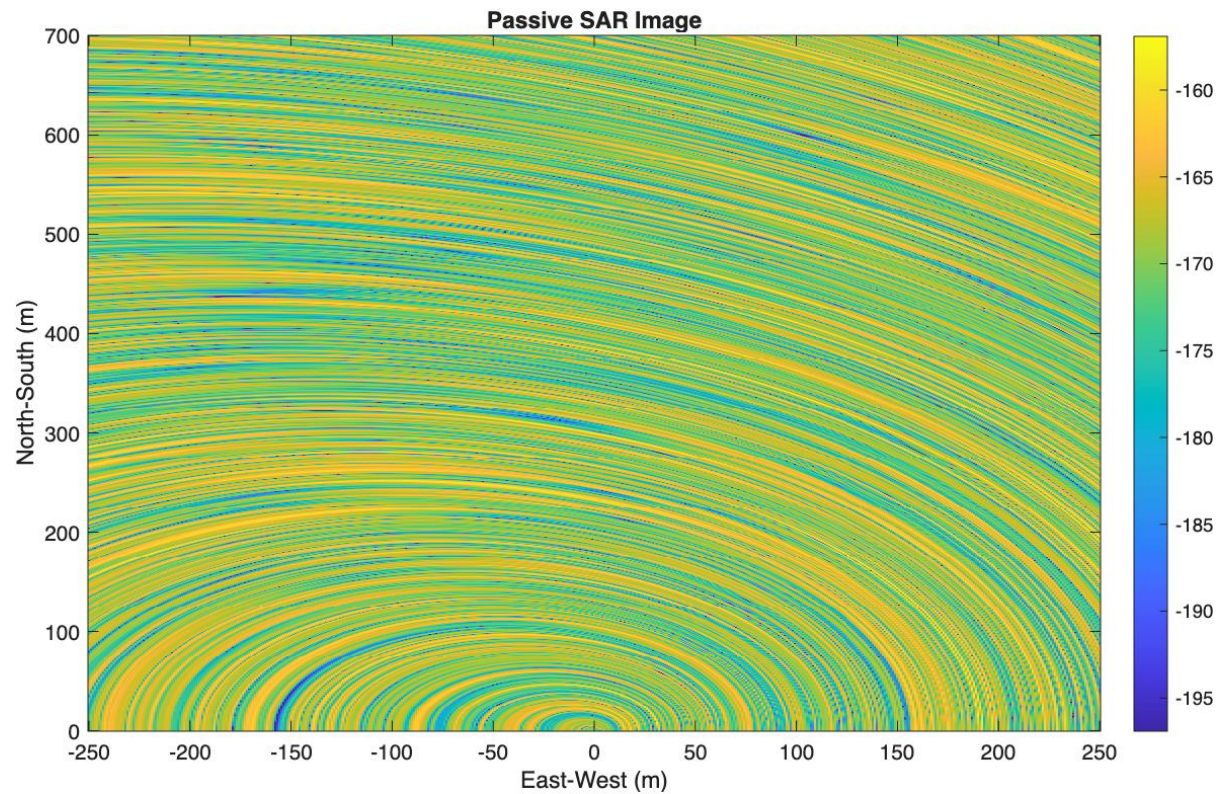
Boston University College of Engineering



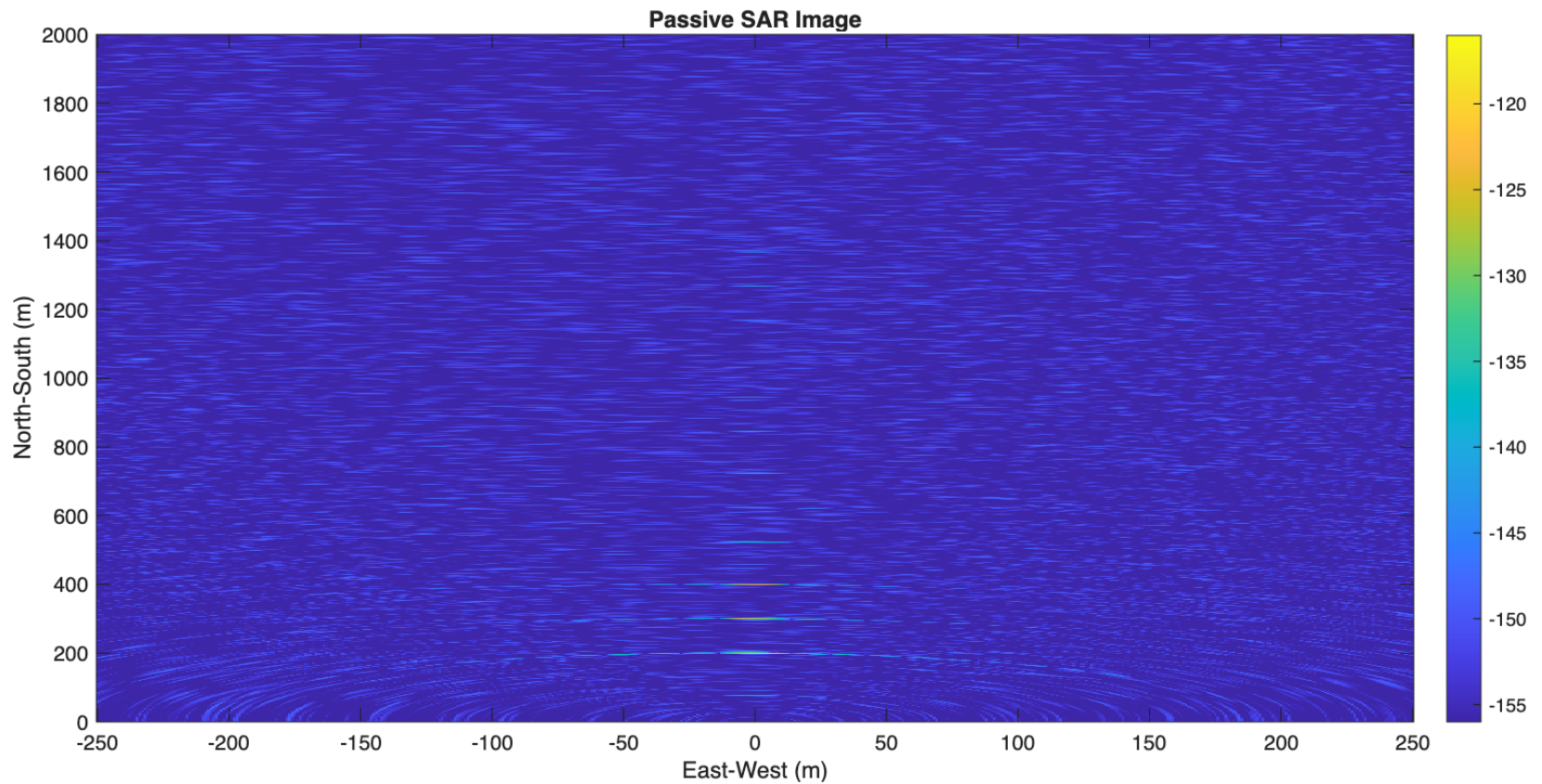
Conclusions and Outlook

- Developed a program to produce Starlink waveforms from arbitrary data files
- Successfully demonstrated the use Starlink waveforms in detecting point targets in an empty scene
- Active research topic
- Future Work
 - Further improvement of the spectrum in the signal generation through better PAPR algorithm
 - Direct path inclusion and mitigation
 - SNR dependent detection mode using SSS sequences
 - Antenna directionality
 - Satellite TLE data
 - Terrain targets, would require an extensive redesign

Additional Figures



High SNR



Ambiguity Function SSS

